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Development of the Thailand district-level Socioeconomic Deprivation Index: a census-based methodological study

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Abstract

Background Area-level socioeconomic deprivation is a key determinant of health disparities and quality of life. This study aimed to develop and validate the Thailand district-level Socioeconomic Deprivation Index (TSDI)—a nation-wide, district-level index to identify deprived areas and their specific deprivation dimensions using national census data.

Methods In this methodological cross-sectional study, we constructed the TSDI from a 20% stratified random sample of the 2010 Thai national census (63 478 990 individuals across 928 districts). Domains and indicators were selected on the basis of a scoping review and theoretical frameworks of relative deprivation. Principal component analysis (PCA) was used to create a composite index, which was visualized through choropleth mapping.

Results The TSDI comprises six principal components encompassing 29 indicators, explaining 73.82% of the total variance. The components represent: (1) overall socioeconomic deprivation, (2) housing and assets, (3) demography and family composition, (4) foreign and elementary workers, (5) household crowding and (6) access to tap water. Spatial analysis revealed severe deprivation clusters in the northern, lower northeastern and southern regions, while central (including Bangkok) and eastern coastal areas were less deprived.

Conclusions The TSDI is a novel, validated tool that provides a comprehensive and spatially explicit measure of socioeconomic deprivation in Thailand. It offers critical data for targeting public health interventions, informing resource allocation and investigating the pathways linking social inequality to health and development outcomes.

Keywords Socioeconomic deprivation index, Small-area analysis, Principal component analysis, Methodological study, Ecological study, Thailand

Background

Socioeconomic deprivation refers to the relative disadvantage in accessing and sustaining economic status, social resources and opportunities [1–4]. It is closely associated with poor quality of life and low life satisfaction [5, 6] and is a well-established determinant of population health and well being [7–9], contributing significantly to health inequalities [10–12]. Recent evidence indicates that individuals residing in socioeconomically deprived areas face higher risks of morbidity [13–16],

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poorer treatment responses [17] and increased mortality [18, 19].

Max Weber's theory of social stratification posits that socioeconomic status reflects hierarchical distinctions shaped by class, status and power. This framework has informed the use of life-chance indicators to identify marginalized or disadvantaged groups [20]. Building on this, Peter Townsend later conceptualized deprivation as a multidimensional phenomenon, emphasizing that it extends beyond economic hardship to encompass broader aspects of living conditions that impair health and wellbeing [1–6].

In practice, socioeconomic deprivation is often assessed using area-level deprivation indices. These composite measures aggregate indicators such as education, income, housing quality, utilities, and demographic characteristics to quantify disadvantage within defined geographical units [4, 21–23]. Over time, methodologies for measuring deprivation have evolved. Early approaches, such as the Foster–Greer–Thorbecke (FGT) index, relied heavily on income or consumption to capture poverty [24–26]. More recent frameworks, including the Alkire–Foster Multidimensional Poverty Index (MPI), integrate nonmonetary dimensions—such as education, health and living standards—to provide a more holistic view of deprivation [27]. Such tools are vital for informing equitable policy design [21], and numerous country-specific deprivation indices have been developed, including in the United States [28–30], the United Kingdom [31, 32], Saudi Arabia [33], China [34, 35], Republic of Korea [36–39], and Thailand [40–44].

Thailand, an upper-middle-income country, exhibits substantial regional socioeconomic disparities. While national-level inequality has decreased in recent years [40–43], and the Universal Coverage Scheme has improved access to health services, significant gaps in health outcomes across socioeconomic groups remain [43]. These disparities have been further exacerbated during the coronavirus disease 2019 (COVID-19) pandemic, underscoring the role of socioeconomic inequality as a key determinant of health [41].

Spatial analyses offer additional insights into how deprivation shapes health outcomes. For example, studies in Thailand have linked urban expansion and smoking behavior to increased population density and accelerated COVID-19 transmission [45]. Similarly, the incidence of colorectal cancer has been associated with high nighttime light intensity—a proxy for urbanization—particularly in Bangkok and its metropolitan region [46]. Such findings highlight the complex pathways through which deprivation, urbanization, and behavioral transitions influence health. Further complexity is illustrated by findings that internet access was positively associated with antibiotic use among older adults, whereas higher physician density was negatively correlated, suggesting that both technological access and health system characteristics shape health behaviors [47].

Effective policy responses to socioeconomic inequality require not only the identification of deprived populations but also an understanding of local contextual factors and the administrative jurisdictions responsible for intervention [48]. In Thailand, poverty measurement has advanced from traditional income-based metrics—developed by the National Statistical Office (NSO) and the National Economic and Social Development Board (NESDB)—to the Thai People Map and Analytics Platform (TPMAP). A collaboration between NESDB and the National Electronics and Computer Technology Center (NECTEC), TPMAP provides a multidimensional assessment of poverty across health, living conditions, education, income and access to services [49].

Despite these advances, existing tools such as TPMAP focus primarily on household-level wellbeing and income poverty. They may not fully capture structural and spatial aspects of deprivation—such as those shaped by demographic change, urbanization, migration and regional inequality—that influence health systems and service access [50]. A more comprehensive, area-based measure is needed to reflect the multidimensional nature of socioeconomic contexts and their impact on health disparities.

Therefore, this study aimed to develop the Thailand district-level Socioeconomic Deprivation Index (TSDI) using the most recent national census data. The TSDI is designed to identify deprived areas and the key dimensions of deprivation, providing local authorities with an evidence-based tool to support targeted social and health policy interventions.

Methods

Development of the Thailand district-level Socioeconomic Deprivation Index (TSDI)

The Thailand district-level Socioeconomic Deprivation Index (TSDI) was developed using established guidelines for constructing small area-level deprivation indices proposed by Noble et al. [4] and Allik et al. [21]. These frameworks outline key principles for creating multidimensional deprivation indices and provide a structured process for their development, including a discussion of methodological considerations [4, 21]. A cross-sectional analysis of national census data was conducted. The construction of the TSDI followed a stepwise process, with variable selection guided by both conceptual relevance and data availability (Fig. 1).

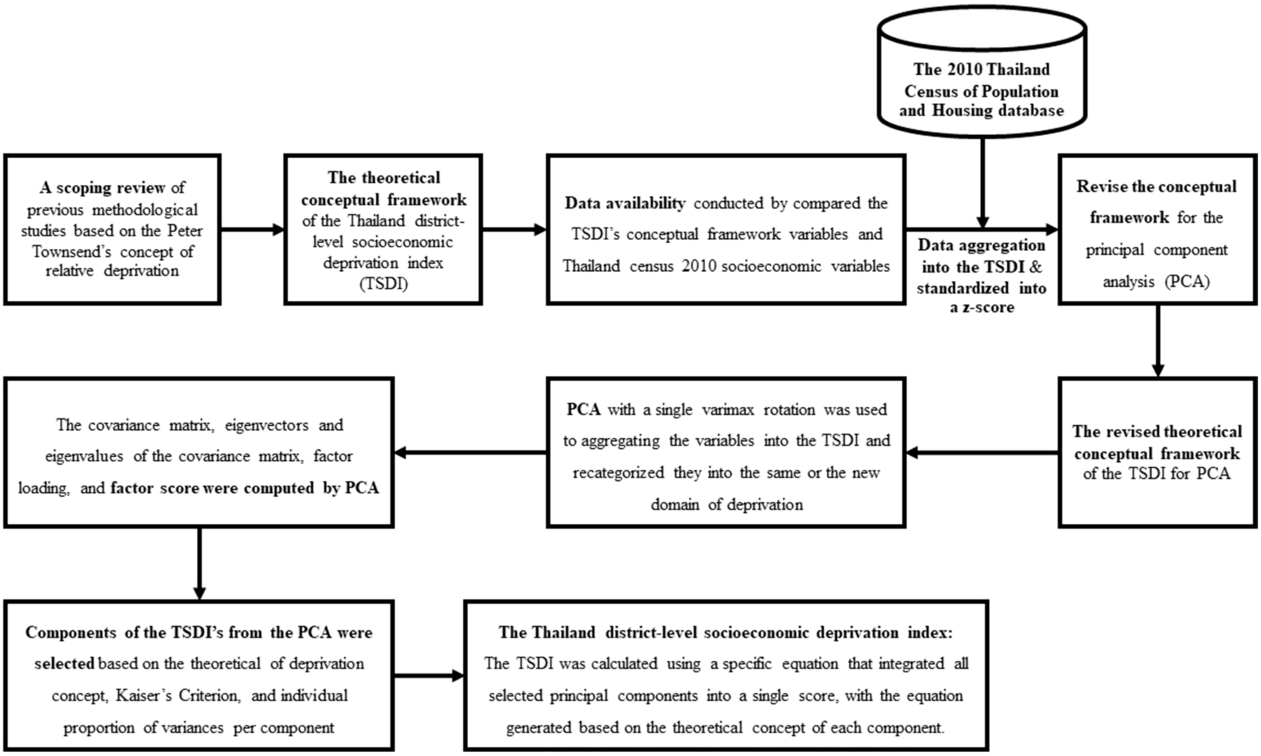

Fig. 1 Methodology flowchart for constructing the Thailand district-level Socioeconomic Deprivation Index (TSDI)

Data source

Data were obtained from the 2010 Census of Population and Housing, administered by the National Statistical Office (NSO) of Thailand. The census is conducted decennially; the 2010 census was the most recent available owing to delays from the COVID-19 pandemic. It collects comprehensive household and individual-level data on demographics, migration, residency duration, dwelling characteristics, tenure and ownership of durable assets. The census does not, however, include data on income or expenditure [51].

A systematic random weighted sample of 20% of the census database—the maximum proportion accessible for research—was provided by the NSO. In 2010, Thailand was classified as a lower-middle-income country with a population of 65.98 million, administratively divided into 928 districts [51].

Conceptual framework and dimension selection

Guided by methodological guidelines [4, 21], we first defined the construct to be measured. The TSDI is grounded in Peter Townsend's concept of relative deprivation [1] and Michael Noble's perspective on multiple deprivation [4]. Socioeconomic deprivation was thus defined as a multidimensional state of relative disadvantage, encompassing inequalities in socioeconomic status, material resources and social opportunities that prevent individuals or groups from achieving a high quality of life [1, 4].

To operationalize this concept, a scoping review of methodological studies on area-level deprivation indices was conducted [33–35, 37, 52–61] (Additional File 1: Table 2). Domains and candidate variables were selected on the basis of their theoretical fit with Townsend's framework [1, 21]. These were subsequently synthesized into a theoretical conceptual framework comprising seven domains and 34 indicators (Fig. 2), with detailed rationale provided in Additional File 2.

The defined domains are:

1. Education: Lack of basic educational attainment, reflecting early life socioeconomic status and impacting social integration.
2. Income, employment and occupation: Inequality related to low income, unstable employment and exclusion from the labor market.
3. Demography and family composition: Proportions of vulnerable population groups (e.g. ethnic minorities, single-parent households) who may face discrimination or limited social support.

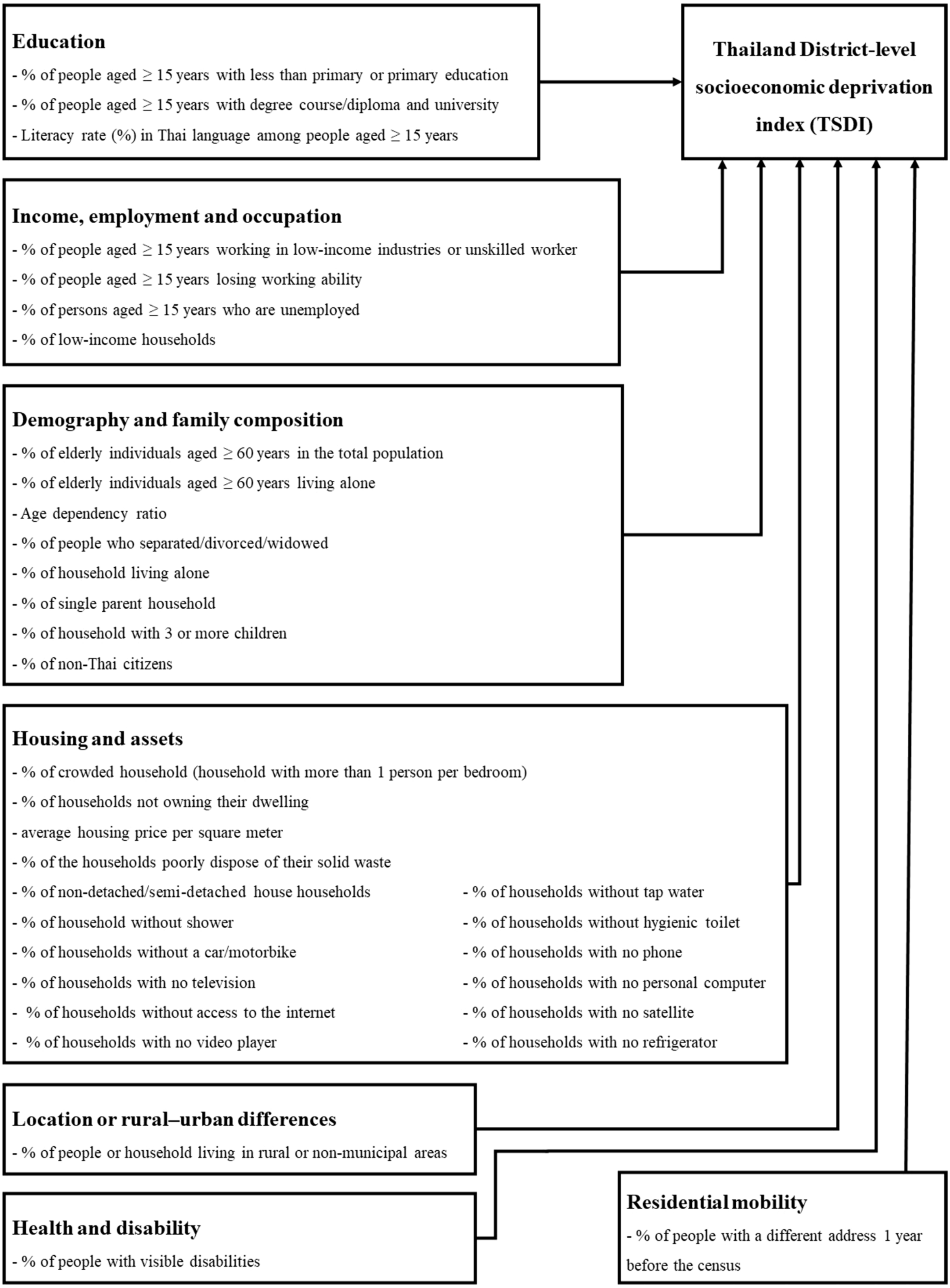

Fig. 2 Theoretical conceptual framework of the TSDI

4. Housing and assets: Inequalities in housing conditions and material living standards.
5. Location or rural–urban differences: Disadvantage associated with living outside municipal areas, implying limited access to infrastructure and services.
6. Health and disability: Risk of impaired quality of life due to poor physical health or disability, affecting social participation and socioeconomic circumstances.
7. Residential mobility: Population churn, reflecting the paradox where less deprived areas may attract more migrants.

Data preparation and aggregation

All variables from the theoretical framework were assessed for availability in the 2010 census. The final list of variables for analysis was determined on the basis of this assessment (Additional File 2: Supplementary Fig. S1). Only data from individual and private households were analyzed. Raw count data were aggregated to the district level as percentages or ratios. Each aggregated variable was then standardized to a z-score to ensure all indicators were on a comparable scale for analysis [21].

Statistical analysis and index construction

Principal component analysis (PCA) was used to construct the TSDI. PCA is a dimensionality reduction technique suitable for combining multiple correlated variables into a composite index while retaining maximum variance from the original data [62]. It is widely used in deprivation index construction and helps avoid double-counting [21].

PCA with varimax rotation was performed on the standardized variables. Component selection was based on conceptual relevance, Kaiser's criterion (eigenvalue ≥ 1), and a minimum proportion of variance explained ($\geq 10\%$ per component). The final TSDI score for each district was calculated via a specific equation that integrated all selected principal components, weighted by their respective variances. Analyses were conducted using IBM SPSS Statistics (version 28.0).

Choropleth mapping

Districts were classified into quintiles on the basis of their TSDI scores, from the least deprived (1st quintile) to the most deprived (5th quintile). A choropleth map was generated using GeoDa software to visualize the geographical distribution of socioeconomic deprivation across Thailand, linking TSDI scores to district shapefiles obtained from the International Health Policy Program (IHPP), Thailand.

Validation of the index

The TSDI was validated across three dimensions, consistent with international guidelines for composite indicators [63] and spatial analysis [64–66].

1. Internal structural validation: The suitability of the data for PCA was assessed using the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity. The retained components were evaluated on the basis of eigenvalues, factor loadings, explained variance and conceptual coherence.
2. Spatial consistency assessment: The geographic distribution of TSDI scores was visually examined via choropleth maps and qualitatively assessed for alignment with well-documented regional socioeconomic patterns in Thailand [42, 67, 68].
3. Empirical convergent validation: Convergent validity was tested by examining the association between TSDI scores and three district-level mortality outcomes with established socioeconomic gradients: all-cause, stroke and chronic obstructive pulmonary disease (COPD) mortality. Mortality data were sourced from the Health Information System Development Office (HISO), Thailand Ministry of Public Health.

Spatial regression was employed to account for spatial autocorrelation. After detecting significant spatial dependence in ordinary least squares (OLS) residuals using Global Moran's I, generalized spatial two-stage least squares (GS2SLS) models were fitted, including spatial lag models (SLM) and spatial error models (SEM). A statistically significant positive association between the TSDI and mortality rates in these spatially adjusted models was considered evidence of convergent validity. These analyses were performed using Stata version 18.0.

Results

Sample characteristics and index construction

The final analytical sample, derived from a 20% random weighted sample of the 2010 national census, included 63 478 990 individuals and 26 878 768 households across all 928 districts of Thailand, after excluding populations in nonresidential areas. The distribution of socioeconomic deprivation variables across these districts is presented in Table 1.

Principal component analysis (PCA) demonstrated strong internal validity for the Thailand district-level Socioeconomic Deprivation Index (TSDI). The dataset was highly suitable for dimension reduction, as indicated by a Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy of 0.860 and a highly significant Bartlett's test

Table 1 Descriptive statistics of the TSDI indicators ($N = 928$ districts)

Domain of deprivation	N	Mean ± SD	Min	Max
Education				
Percentage of people ≥ 15 years with less than primary or primary education	928	49.5 ± 9.09	15.3	67.9
Percentage of people ≥ 15 years with degree course, diploma and university	928	9.4 ± 7.78	2.4	48.2
Literacy rate (%) among people ≥ 15 years	928	96.8 ± 4.88	62.6	100.0
Income, employment and occupation				
Percentage of people ≥ 15 years working in elementary occupations	928	5.4 ± 4.08	0.0	34.3
Percentage of persons aged over 15 years who are unemployed	928	11.4 ± 3.81	0.9	27.5
Demography and family composition				
Percentage of elderly individuals ≥ 60 years	928	14.1 ± 3.40	4.2	24.3
Percentage of elderly individuals ≥ 60 years living alone	928	1.3 ± 0.53	0.2	3.7
Age dependency ratio per 100 persons of working age[a]	928	56.45 ± 12.65	16.28	90.84
Percentage of separated, divorced or widowed individuals	928	11.5 ± 2.21	3.0	17.1
Percentage of households living alone	928	14.4 ± 5.12	5.0	50.9
Percentage of households with 3 or more children	928	12.0 ± 3.86	2.6	22.5
Percentage of non-Thai citizens	928	2.3 ± 6.00	0.0	77.7
Housing and assets				
Percentage with crowded households	928	58.2 ± 14.73	30.2	85.9
Percentage of households not owning their dwelling	928	11.8 ± 15.07	0.5	83.7
Percentage of nondetached-/semidetached-house households	928	10.5 ± 15.97	0.1	92.0
Percentage of households without tap water	928	23.3 ± 19.87	0.0	91.3%
Percentage of households without hygienic toilet	928	0.4 ± 1.71	0.0	28.9
Percentage of households without a car/motorbike for commuting	928	15.7 ± 10.54	3.0	77.6
Percentage of households with no phone	928	11.6 ± 6.77	1.4	61.5
Percentage of households with no television	928	5.3 ± 6.35	0.4	60.1
Percentage of households with no personal computer	928	78.8 ± 10.13	34.4	95.5
Percentage of households without access to the internet	928	92.7 ± 8.33	45.1	100.0
Percentage of households with no satellite	928	64.8 ± 19.99	13.2	98.2
Percentage of households with no video player	928	28.6 ± 7.22	8.3	72.1
Percentage of households with no refrigerator	928	14.8 ± 9.88	2.4	81.0
Location or rural–urban differences				
Percentage of household in rural/nonmunicipal areas	928	67.4 ± 27.72	0.0	100.0
Health and disability domain				
Percentage of persons with visible disabilities	928	1.0 ± 0.41	0.1	2.6
Residential mobility				
Percentage of persons with a different address 1 year before the census	928	1.2 ± 1.10	0.0	9.6

Data represent the mean percentage ± standard deviation (SD) [a]Persons per 100 persons of working age

of sphericity ($P < 0.001$). Six principal components (PCs) with eigenvalues ≥ 1 were retained, collectively explaining 73.82% of the total variance. This high cumulative variance indicates a stable underlying multidimensional structure appropriate for a composite deprivation measure. The eigenvalues for the six components were 9.10, 4.67, 2.46, 1.86, 1.46 and 1.12, respectively.

Dimensions of socioeconomic deprivation

The factor loadings for each variable are detailed in Table 2. The six retained components were interpreted as distinct dimensions of deprivation:

– PC1 (overall socioeconomic deprivation): Accounting for 32.51% of the variance, this component loaded

Table 2 Factor loadings for the indicators of the TSDI

Domain of deprivation	Factor loading from PC					
	1st	2nd	3rd	4th	5th	6th
Education						
Percentage of people ≥ 15 years with less than primary or primary education	0.846	0.022	0.323	0.102	−0.164	0.024
Percentage of people ≥ 15 years with degree course, diploma and university	−0.947	0.027	−0.151	−0.005	0.020	−0.113
Literacy rate (%) among people ≥ 15 years	0.004	−0.493	0.129	−0.633	0.126	−0.205
Income, employment and occupation						
Percentage of people ≥ 15 years working in elementary occupations	−0.119	−0.096	0.195	0.650	0.275	−0.044
Percentage of persons aged over 15 years who are unemployed	−0.756	−0.114	0.054	0.016	0.105	0.323
Demography and family composition						
Percentage of elderly individuals ≥ 60 years	0.169	−0.143	0.861	−0.196	−0.094	−0.149
Percentage of elderly individuals ≥ 60 years living alone	0.045	−0.006	0.895	0.081	0.139	−0.045
Age dependency ratio per 100 persons of working age	0.699	0.118	0.381	−0.411	−0.059	0.051
Percentage of separated, divorced or widowed individuals	0.195	−0.235	0.789	0.052	0.086	0.109
Percentage of households living alone	−0.556	−0.015	0.138	0.339	0.507	0.019
Percentage of household with 3 or more children	0.589	−0.071	0.188	−0.499	0.147	0.134
Percentage of non-Thai citizens	−0.175	0.286	−0.239	0.655	0.028	−0.052
Housing and assets						
Percentage with crowded households	−0.122	−0.086	−0.041	−0.128	−0.837	0.079
Percentage of households not owning their dwelling	−0.732	0.100	−0.320	0.254	0.358	−0.218
Percentage of nondetached-/semidetached-house households	−0.752	0.088	−0.303	0.199	0.276	−0.265
Percentage of households without tap water	0.219	0.173	−0.134	0.009	−0.020	0.790
Percentage of households without hygienic toilet	0.112	0.683	−0.014	0.074	−0.086	0.135
Percentage of households without a car/motorbike for commuting	−0.374	0.546	−0.123	0.02	0.280	−0.488
Percentage of households with no phone	0.058	0.901	0.107	0.051	−0.063	−0.032
Percentage of households with no television	−0.145	0.888	−0.189	0.127	0.016	−0.060
Percentage of households with no personal computer	0.927	0.139	0.005	−0.083	0.130	0.033
Percentage of households without access to the internet	0.908	−0.003	0.157	−0.026	0.066	0.143
Percentage of households with no satellite	−0.351	0.080	−0.092	−0.352	0.423	0.071
Percentage of households with no video player	0.199	0.832	0.055	−0.061	0.159	0.103
Percentage of households with no refrigerator	−0.058	0.871	−0.292	0.037	0.155	−0.040
Location or rural–urban differences						
Percentage of household in rural/nonmunicipal areas	0.722	−0.030	0.007	−0.064	0.091	0.288
Health and disability domain						
Percentage of persons with visible disabilities	0.321	0.068	0.540	−0.142	−0.147	−0.045
Residential mobility						
Percentage of persons with a different address 1 year before the census	−0.490	−0.052	−0.186	0.435	0.309	0.166

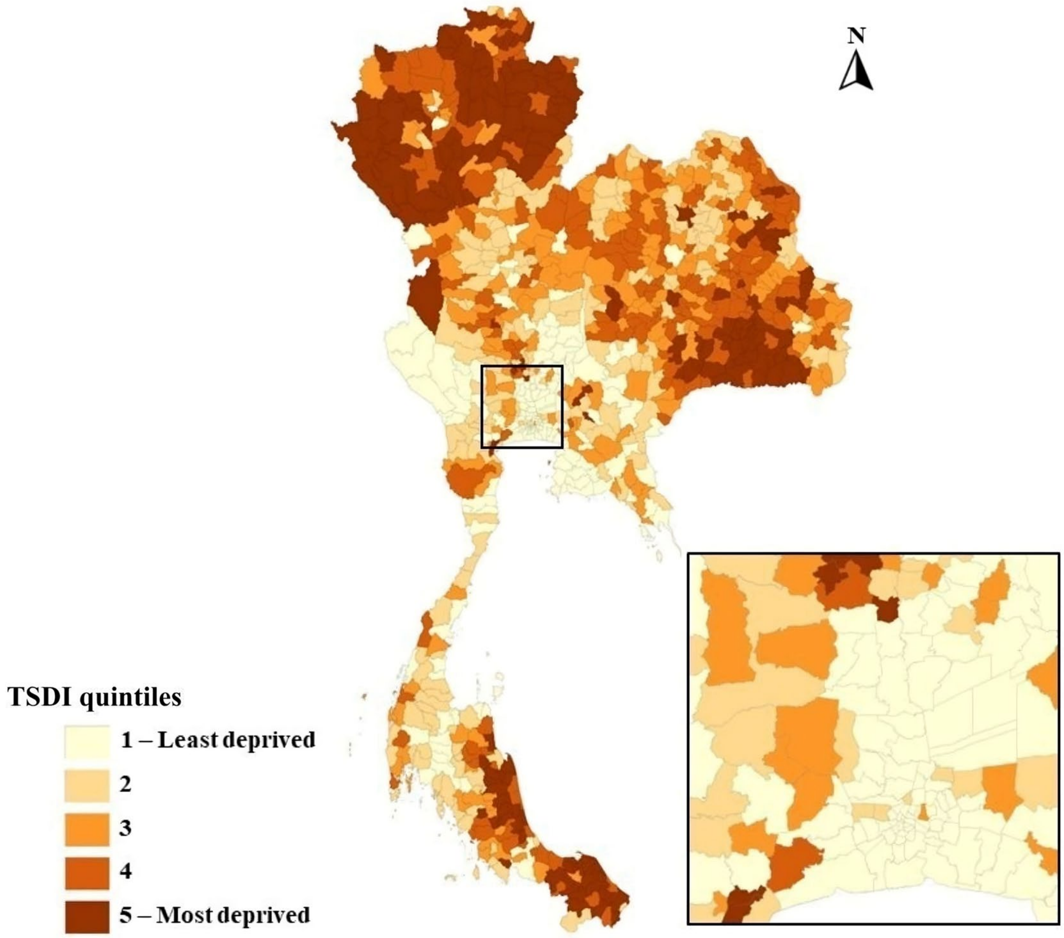

Fig. 3 Spatial distribution of the Thailand district-level Socioeconomic Deprivation Index (TSDI) by district quintiles

highly on a broad range of indicators encompassing education, employment, demography, housing and location, representing a general deprivation factor.

- PC2 (housing and assets): Explaining 16.67% of the variance, this component was defined by indicators of material deprivation related to housing conditions and asset ownership.
- PC3 (demography and family composition): This component (8.78% of variance) was characterized by variables indicating vulnerable demographic structures, such as older age, living alone and being divorced/separated/widowed.
- PC4 (foreigner and elementary workers): Accounting for 6.65% of the variance, this component combined indicators of foreign national presence and employment in elementary occupations.
- PC5 (crowdedness): This component (5.22% of variance) was defined by variables related to large family size and household crowding.
- PC6 (tap water): Explaining 3.99% of the variance, this component was primarily defined by the lack of access to tap water.

To ensure all components contributed coherently to the final deprivation score (where higher values indicate greater deprivation), the factor scores of PC4 and PC5 were inverted (multiplied by −1). The final TSDI score for each district was calculated as an average of the six retained components (1), resulting in a standardized deprivation score for all 928 districts (Table 1 and Table 2).

$$[1st \cdot PC + 2nd \cdot PC + 3rd \cdot PC + (-1 * \cdot 4th \cdot PC) + (-1 * \cdot 5th \cdot PC) + 6th \cdot PC] / 6 \tag{1}$$

Choropleth mapping for the Thailand district-level Socioeconomic Deprivation Index

Spatial distribution of deprivation

The geographic distribution of the TSDI, visualized via choropleth mapping, revealed clear regional socioeconomic disparities (Fig. 3). Districts were classified into quintiles, with the 1st quintile representing the least deprived and the 5th the most deprived.

Spatial analysis identified severe deprivation clusters in the northern region, the lower northeastern region (Thai–Cambodia border), and the lower southern region (Gulf of Thailand). In contrast, central Thailand (Bangkok and surrounding areas) and eastern coastal areas were markedly less deprived. This pattern aligns with well-documented regional inequalities in Thailand [42, 67, 68], supporting the geographic validity of the index. The spatial distribution of each individual principal component is further illustrated in Fig. 4.

Empirical convergent validation

The TSDI demonstrated consistent empirical convergent validity through its significant associations with district-level mortality outcomes. Initial ordinary least squares (OLS) regression models showed substantial residual spatial autocorrelation for all-cause, stroke, and COPD mortality (Global Moran's *I*: 390.95–1156.25, all $P<0.001$), necessitating the use of spatial regression models.

Spatial regression revealed robust, positive associations between the TSDI and mortality after adjusting for spatial dependence (Fig. 5).

All-cause mortality: A significant positive association was found in both the spatial lag model (SLM: $\beta=30.21$, $P=0.002$) and the spatial error model (SEM: $\beta=72.00$, $P<0.001$), with the SEM providing a superior model fit.

Stroke mortality: A significant association was identified in the SEM ($\beta=6.59$, $P=0.003$) but not in the SLM ($P=0.085$), suggesting that the relationship is

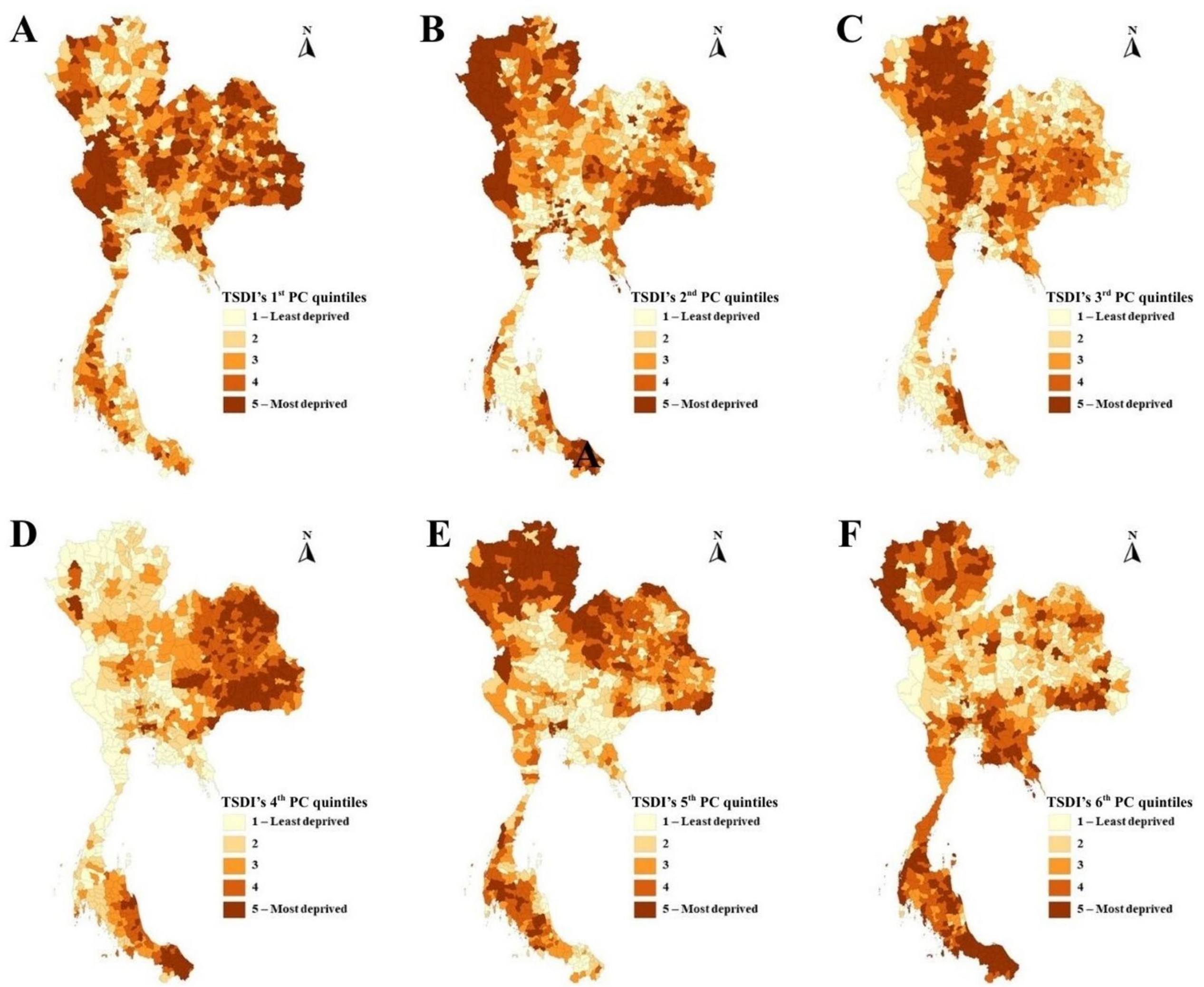

Fig. 4 Spatial distribution of the six principal components of the TSDI

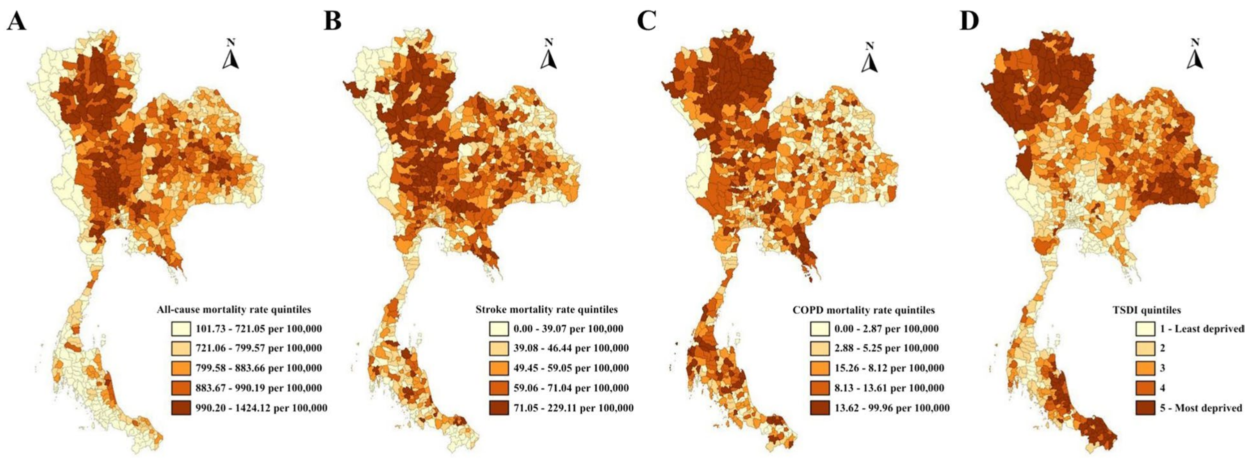

Fig. 5 Spatial distribution at the district-level. **A** All-cause mortality, **B** stroke mortality, **C** COPD mortality and **D** TSDI quintiles

predominantly influenced by spatially correlated unobserved variables rather than spatial spillover effects.

COPD mortality: A significant association was also found in the SEM ($\beta = 2.24$, $P = 0.011$), while the SLM was nonsignificant ($P = 0.224$), indicating a similar pattern of spatial error dependence.

Spatial error dependency was significant across all models ($\lambda = 0.65$–0.86, all $P < 0.001$), and the SEM specification effectively minimized residual spatial autocorrelation. These results confirm that higher socioeconomic deprivation, as measured by the TSDI, is consistently associated with higher district-level mortality risks, validating the index's utility for capturing health-relevant socioeconomic gradients.

Discussion

This study successfully developed the Thailand district-level Socioeconomic Deprivation Index (TSDI), a nationwide, population-based, small-area index that addresses a significant gap in deprivation measures for Southeast Asia. While numerous such indices exist for high-income countries [33–35, 37, 52–61], comparable tools for lower-middle-income nations in this region have been limited. Constructed from Thailand's decennial national census [51], the TSDI provides a single, standardized scale derived via principal component analysis (PCA), enhancing its communicability [62] and suitability for studies on health and socioeconomic inequality [21]. The use of choropleth mapping further augments its utility, enabling the intuitive visualization of complex spatial patterns and disparities across the country [69–71]. Although not the first deprivation index developed for Thailand, the TSDI incorporates a broader range of components than previous efforts, offering a more comprehensive reflection of multidimensional deprivation, particularly for investigating health outcomes and inequality [44].

The TSDI in context: a multidimensional tool for public health

The district-level TSDI represents a critical advancement for addressing the socioeconomic determinants of health in Thailand. Urban growth and population concentration exacerbate social inequality and directly influence disease patterns, including increased susceptibility to communicable diseases in high-density settings and shifting risk profiles for noncommunicable diseases [72, 73]. By integrating multidimensional indicators—encompassing education; income and employment; demography; housing and assets; location; health and disability; and residential mobility—the TSDI provides a holistic picture of deprivation [4, 21]. This comprehensive approach facilitates the accurate identification of areas most in need of public health interventions [29, 74, 75] and enables robust analysis of the links between deprivation and health outcomes, from disease transmission to medication use [43, 76–78]. Consequently, the TSDI is a vital tool for formulating equitable public health policies and guiding resource allocation, ultimately contributing to reduced inequality and more effective, sustainable healthcare for all population groups [79, 80].

Comparative performance and theoretical foundation

The TSDI accounted for 73.82% of the total variance in area deprivation, a robust result that sits within the spectrum of international indices—higher than those for Cyprus (65.7%) [37] and Canada (58.9%) [53], though slightly lower than indices for Saudi Arabia (80.98%) [33] and China (81.1%) [34]. This confirms that the six selected components effectively represent socioeconomic status in Thailand. The index's development was grounded in the theoretical frameworks of Peter Townsend's relative deprivation [1] and Michael Noble's multiple deprivation [4], ensuring conceptual rigor from the outset, as recommended by leading methodologies [4, 21]. This theoretical foundation, combined with a comprehensive review of international indicators [33–35, 37, 52–59, 61], resulted in a final structure of six principal components that capture all theoretical dimensions of deprivation [1–6].

The TSDI demonstrates strong alignment with established indices such as the Townsend index [22] through shared indicators such as unemployment and crowdedness. Furthermore, its identification of deprived regions corresponds closely with national reports on poverty and inequality [81–83]. A key advantage of the TSDI over binary classifications is its provision of a continuous, standardized measure of deprivation for each district, enabling more precise targeting of interventions. By incorporating both material and social deprivation markers, including those relevant to rural areas [84, 85], the TSDI addresses a known weakness of traditional indices that focus solely on material deprivation and often show weaker associations with health in rural settings [86–88].

Spatial patterns and regional development

The spatial distribution of the TSDI reveals stark regional inequalities that reflect Thailand's economic geography. Severe deprivation is concentrated in the northern, lower northeastern (Thai–Cambodia border), and lower southern regions (Gulf of Thailand), while the central region (Bangkok and environs) and eastern coastal areas are less deprived. This pattern is consistent with long-standing disparities documented in national and international assessments, supporting the geographic validity of the index [42, 67, 68].

These spatial patterns are deeply intertwined with national development policies. The lower deprivation in the eastern region can be attributed to decades of targeted industrialization, beginning with the Eastern Seaboard Development Program in the 1980s and continuing with the recent Eastern Economic Corridor (EEC) project [89]. These initiatives fostered advanced manufacturing and infrastructure, pulling economic activity and reducing deprivation in these zones. A space–time analysis corroborates this, showing that the multidimensional poverty index increases with distance from industrial parks [90]. Similarly, the relative affluence of Songkhla in the otherwise deprived south may reflect the success of a shift to holistic, "people-centered development" in the Songkhla Lake Basin [91].

The index's components are sensitive to these rural–urban dynamics. The high loading of the "foreigner and elementary worker" component in economic zones reflects labor demand, while components such as "housing and assets," "crowdedness," and "tap water" clearly delineate deprivation in the north, northeast and south. The inclusion of "recent immigration" and "health deprivation"—rare in other indices [86, 89]—further enhances the TSDI's uniqueness and comprehensiveness.

Validation and limitations

The TSDI demonstrates strong validity. Internal structural validation was supported by a high KMO statistic and significant Bartlett's test, consistent with best practices for composite indicators [63]. The spatial distribution of scores aligned with known regional disparities, confirming geographic consistency [42, 67, 68]. Finally, empirical convergent validity was established through significant, spatially adjusted associations with all-cause, stroke and COPD mortality. The need for spatial regression models, indicated by significant Global Moran's I statistics [64], and the subsequent robust findings are consistent with principles of spatial econometrics [65], affirming the index's epidemiological relevance.

This study has several limitations. First, the reliance on the 2010 census introduces a considerable time lag, exacerbated by the COVID-19 pandemic. Second, the index is constrained by the available census variables and does not include direct measures of environmental deprivation or public facility access. This study provides a practical, evidence-based tool for identifying vulnerable areas based on key social determinants of health.

Conclusions

This study developed a robust, validated national socioeconomic deprivation index for Thailand using census data. The TSDI provides an alternative and comprehensive method for examining socioeconomic inequality, using aggregated geospatial analysis to identify deprived areas and their specific deprivation dimensions. This new index holds significant potential for enabling a thorough assessment of the relationship between socioeconomic deprivation and various health outcomes and inequalities in Thailand. With demonstrated internal, spatial and empirical convergent validity, the TSDI is a reliable tool poised to inform health research, inequality monitoring and geographically targeted policy interventions aimed at reducing socioeconomic and health disparities across the nation.

Abbreviations

EEC	Eastern Economic Corridor
ESB	Eastern Seaboard Development Program
MPI	Multidimensional Poverty Index
PC	Principal component
PCA	Principal component analysis
TSDI	Thailand district-level Socioeconomic Deprivation Index

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Author contributions

W.R. was the principal investigator of the study. W.R. initiated the study, retrieved the articles, reviewed the retrieved articles, performed the theoretical synthesis, conducted the data curation, analyzed the data, performed validation, conducted discussions and drafted the manuscript. M.T. guided the research idea and finalized the protocol, conducted the funding acquisition, reviewed the retrieved articles and data analyses, and reviewed and edited the manuscript. K.B. conducted the funding acquisition, advised on the data analyses and reviewed the manuscript. A.S. advised on the data analyses and reviewed the manuscript. All the authors have read and approved the final submitted version of the manuscript.

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Data availability

The original articles included in the scoping review for this methodological study are publicly available. The complete search strategy is available in Additional File 1. Data availability was assessed by comparing the variables of the TSDI's conceptual framework with those from the 2010 Thailand census socioeconomic database. The complete data availability is available in Additional File 2.

Declarations

Ethics approval and consent to participate

This study used existing aggregated data at the district level and could not track individuals. We were unable to request consent from individuals.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

1. Townsend P. Poverty in the United Kingdom: a survey of household resources and standards of living. London: Allen Lane and Penguin Books; 1979.
2. Singh-Manoux A, Marmot MG, Adler NE. Does subjective social status predict health and change in health status better than objective status? Psychosom Med. 2005;67(6):855–61.
3. Ravallion M. Mashup indices of development. World Bank Res Obs. 2012;27(1):1–32.
4. Noble M, Wright G, Smith G, Dibben C. Measuring multiple deprivation at the small-area level. Environ Plann Econom Space. 2006;38(1):169–85.
5. Pettigrew TF. Samuel Stouffer and relative deprivation. Soc Psychol Q. 2015;78(1):7–24.
6. Sgrò A, Cambridge WA, McLean KA, Drake TM, Camilleri-Brennan J, Knight SR, et al. Is socioeconomic deprivation associated with worse quality of life, anxiety and depression in liver transplant recipients? A cross-sectional study in a national transplantation programme. BMJ Open. 2023;13(8):e070422.
7. He Q, Tong H, Liu JB. How does inequality affect the residents' subjective well-being: inequality of opportunity and inequality of effort. Front Psychol. 2022;13:843854.
8. Ugur ZB. How does inequality hamper subjective well-being? The role of fairness. Soc Indic Res. 2021;158(2):377–407.
9. Buttrick NR, Heintzelman SJ, Oishi S. Inequality and well-being. Curr Opin Psychol. 2017;18:15–20.
10. Diez Roux AV. Investigating neighborhood and area effects on health. Am J Public Health. 2001;91(11):1783–9.
11. Hosseinpoor AR, Stewart Williams JA, Itani L, Chatterji S. Socioeconomic inequality in domains of health: results from the World Health Surveys. BMC Public Health. 2012;12:198.
12. Dirksen J, Pinilla-Roncancio M, Wehrmeister FC, Ferreira LZ, Vidaletti LP, Kirkby K, et al. Exploring the potential for a new measure of socioeconomic deprivation status to monitor health inequality. Int J Equity Health. 2022;21(1):56.
13. Wu VS, Caputo FJ, Quatromoni JG, Kirksey L, Lyden SP, Rowse JW. Association between socioeconomic deprivation and presentation with a ruptured abdominal aortic aneurysm. J Vasc Surg. 2024;79(1):44–54.
14. Jansen L, Schwettmann L, Behr C, Eberle A, Holleczek B, Justenhoven C, et al. Trends in cancer incidence by socioeconomic deprivation in Germany in 2007 to 2018: an ecological registry-based study. Int J Cancer. 2023;153(10):1784–96.
15. Jeriček Klanšček H, Furman L. Socioeconomic deprivation and inequalities in mental well-being during the COVID-19 pandemic among adolescents. Int J Environ Res Public Health. 2023. https://doi.org/10.3390/ijerph20136233.
16. Buczkowska M, Butt S, Jenkins C, Hungerford D, Hawker J, Verlander NQ, et al. Association between socioeconomic deprivation and incidence of infectious intestinal disease by pathogen and linked transmission route: an ecological analysis in the UK. Epidemiol Infect. 2023;151:e109.
17. Zhao SS, Rogers K, Kearsley-Fleet L, Watson K, Bosworth A, Galloway J, et al. Socioeconomic deprivation is associated with reduced response and lower treatment persistence with TNF inhibitors in rheumatoid arthritis. Rheumatology (Oxford). 2024;63(3):648–56.
18. Dunlop HM, Williams LJ, Hall PS, Barber M, Dodds C, Figueroa JD. Evaluation of the association of area-level socioeconomic deprivation and breast cancer recurrence by estrogen receptor subtypes in Scotland. Breast Cancer Res. 2023;25(1):106.
19. Lusk JB, Blass B, Mahoney H, Hoffman MN, Clark AG, Bae J, et al. Neighborhood socioeconomic deprivation, healthcare access, and 30-day mortality and readmission after sepsis or critical illness: findings from a nationwide study. Crit Care. 2023;27(1):287.
20. Lynch JW, Kaplan G. Socioeconomic position. In: Berkman LF, Kawachi I, editors. Social Epidemiology. New York: Oxford University Press; 2000. p. 13–35.
21. Allik M, Leyland A, Travassos Ichihara MY, Dundas R. Creating small-area deprivation indices: a guide for stages and options. J Epidemiol Community Health. 2020;74(1):20–5.
22. Townsend P. Deprivation. J Soc Policy. 1987;16(2):125–46.
23. Nolan B, Whelan CT. Resources, deprivation and poverty. Oxford: Clarendon Press; 1996.
24. Caranci N, Biggeri A, Grisotto L, Pacelli B, Spadea T, Costa G. The Italian deprivation index at census block level: definition, description and association with general mortality. Epidemiol Prev. 2010;34(4):167–76.
25. Woods LM, Rachet B, Coleman MP. Choice of geographic unit influences socioeconomic inequalities in breast cancer survival. Br J Cancer. 2005;92(7):1279–82.
26. Foster J, Greer J, Thorbecke E. A class of decomposable poverty measures. Econometrica. 1984;52(3):761–6.
27. Alkire S, Foster J. Counting and multidimensional poverty measurement. J Public Econ. 2011;95(7–8):476–87.

28. Kind AJ, Jencks S, Brock J, Yu M, Bartels C, Ehlenbach W, et al. Neighborhood socioeconomic disadvantage and 30-day rehospitalization: a retrospective cohort study. Ann Intern Med. 2014;161(11):765–74.
29. Kind AJH, Buckingham WR. Making neighborhood-disadvantage metrics accessible - The neighborhood atlas. N Engl J Med. 2018;378(26):2456–8.
30. The Scottish Government. SIMD 2020 technical notes [Internet]. Edinburgh: Scottish Government. 2020. https://www.gov.scot/publications/simd-2020-technical-notes/. Accessed 22 Aug 2023
31. Jordan H, Roderick P, Martin D. The Index of Multiple Deprivation 2000 and accessibility effects on health. J Epidemiol Community Health. 2004;58(3):250–7.
32. McLennan D, Noble S, Noble M, Plunkett E, Wright G, Gutacker N. English indices of deprivation 2019: technical report. London: Ministry of Housing, Communities & Local Government; 2019.
33. AlOmar RS, Parslow RC, Law GR. Development of two socioeconomic indices for Saudi Arabia. BMC Public Health. 2018;18(1):791.
34. Wang K, Law CK, Zhao J, Hui AY, Yip BH, Yeoh EK, et al. Measuring health-related social deprivation in small areas: development of an index and examination of its association with cancer mortality. Int J Equity Health. 2021;20(1):216.
35. Wang Z, Chan KY, Poon AN, Homma K, Guo Y. Construction of an area-deprivation index for 2869 counties in China: a census-based approach. J Epidemiol Community Health. 2021;75(2):114–9.
36. Panczak R, Berlin C, Voorpostel M, Zwahlen M, Egger M. The Swiss neighborhood index of socioeconomic position: update and revalidation. Swiss Med Wkly. 2023;153:40028.
37. Lamnisos D, Lambrianidou G, Middleton N. Small-area socioeconomic deprivation indices in Cyprus: development and association with premature mortality. BMC Public Health. 2019;19(1):627.
38. Pampalon R, Hamel D, Gamache P, Raymond G. A deprivation index for health planning in Canada. Chronic Dis Can. 2009;29(4):178–91.
39. Khodayari Moez E, Maximova K, Sim S, Senthilselvan A, Pabayo R. Developing a socioeconomic status index for chronic disease prevention research in Canada. Int J Environ Res Public Health. 2022. https://doi.org/10.3390/ijerph19137800.
40. Cook S, Pincus J. Poverty, inequality and social protection in Southeast Asia: an introduction. Southeast Asian Econ. 2014;31:1–17.
41. Bird K, Hattel K, Sasaki E, Attapich L. Poverty, income inequality, and microfinance in Thailand. Manila: Asian Development Bank. Mandaluyong; 2011.
42. World Bank Group. Thailand rural income diagnostic: challenges and opportunities for rural farmers. Bangkok: World Bank; 2022.
43. Kaikeaw S, Punpuing S, Chamchan C, Prasartkul P. Socioeconomic inequalities in health outcomes among Thai older population in the era of universal health coverage: trends and decomposition analysis. Int J Equity Health. 2023;22(1):144.
44. Aungkulanon S, Tangcharoensathien V, Shibuya K, Bundhamcharoen K, Chongsuvivatwong V. Area-level socioeconomic deprivation and mortality differentials in Thailand: results from principal component analysis and cluster analysis. Int J Equity Health. 2017;16(1):117.
45. Yotha N, Phimha S, Prasit N, Senahad N, Sirikarn P, Nonthamat A. Spatial association patterns with cultural and behaviour with the situations of COVID-19. Int J Geoinf. 2023;19(4):51–63.
46. Monkhan N, Phimha S, Prasit N, Senahad N. Spatial distribution patterns of colorectal cancer patients in Thailand. Internat J Geoinformat. 2025;21(2):159.
47. Prasit N, Phimha S, Nonthamat A, Nilnate N, Nidthumsakul N, Sresutham P. The impact of health and technology shifts on antibiotic use among the elderly in Thailand. Sci Rep. 2025;15:6220.
48. Herd P, Hoynes H, Michener J, Moynihan D. Introduction: administrative burden as a mechanism of inequality in policy implementation. RSF. 2023;9(5):1–30.
49. Thai People Map and Analytics Platform (TPMAP). About TPMAP. 2019. https://www.tpmap.in.th/about. Accessed 25 May 2020.
50. Mah JC, Penwarden JL, Pott H, Theou O, Andrew MK. Social vulnerability indices: a scoping review. BMC Public Health. 2023;23(1):1253.
51. National Statistical Office. The 2010 population and housing census: Bangkok. Bangkok: National Statistical Office; 2010.
52. Morales K, Flores M, Mendez Y. Analysis of principal nonlinear components for the construction of a socioeconomic stratification index in Ecuador. Rev Desarro Soc. 2021;88:43–82.
53. Ribeiro AI, Launay L, Guillaume E, Launoy G, Barros H. The Portuguese version of the European Deprivation Index: development and association with all-cause mortality. PLoS ONE. 2018;13(12):e0208320.
54. Guillaume E, Pornet C, Dejardin O, Launay L, Lillini R, Vercelli M, et al. Development of a cross-cultural deprivation index in five European countries. J Epidemiol Community Health. 2016;70(5):493–9.
55. Chan E, Serrano J, Chen L, Stieb DM, Jerrett M, Osornio-Vargas A. Development of a Canadian socioeconomic status index for the study of health outcomes related to environmental pollution. BMC Public Health. 2015;15:714.
56. Yuan Y, Wu F. The development of the index of multiple deprivations from small-area population census in the city of Guangzhou, PRC. Habitat Int. 2014;41:142–9.
57. Panczak R, Galobardes B, Voorpostel M, Spoerri A, Zwahlen M, Egger M. A swiss neighborhood index of socioeconomic position: development and association with mortality. J Epidemiol Community Health. 2012;66(12):1129–36.
58. Pornet C, Delpierre C, Dejardin O, Grosclaude P, Launay L, Guittet L, et al. Construction of an adaptable European transnational ecological deprivation index: the French version. J Epidemiol Community Health. 2012;66(11):982–9.
59. Juhász A, Nagy C, Páldy A, Beale L. Development of a deprivation index and its relation to premature mortality due to diseases of the circulatory system in Hungary, 1998-2004. Soc Sci Med. 2010;70(9):1342–9.
60. Sanusi YA. Application of human development index to measurement of deprivations among urban households in Minna, Nigeria. Habitat Int. 2008;32(3):384–98.
61. Messer LC, Laraia BA, Kaufman JS, Eyster J, Holzman C, Culhane J, et al. The development of a standardized neighborhood deprivation index. J Urban Health. 2006;83(6):1041–62.
62. Jolliffe IT, Cadima J. Principal component analysis: a review and recent developments. Philos Trans R Soc Lond A Math Phys Eng Sci. 2016;374(2065):20150202.
63. OECD. Handbook on constructing composite indicators: methodology and user guide. Paris: OECD Publishing; 2008.
64. Moran PAP. Notes on continuous stochastic phenomena. Biometrika. 1950;37(1–2):17–23.
65. Anselin L. Spatial econometrics: methods and models. Dordrecht: Kluwer Academic Publishers; 1988.
66. Waller LA, Gotway CA. Applied spatial statistics for public health data. Hoboken: Wiley; 2004.
67. Phulkerd S, Thongcharoenchupong N, Chamratrithirong A, Gray RS, Pattaravanich U, Ungchusak C, et al. Socio-demographic and geographic disparities of population-level food insecurity during the COVID-19 pandemic in Thailand. Front Public Health. 2023;10:1071814.
68. OECD; Office of the National Economic and Social Development Council (NESDC). Multi-dimensional country review of Thailand. Paris: OECD Publishing; 2019.
69. Crampton J. GIS and geographic governance: reconstructing the choropleth map. Cartographica. 2004;39(1):41–53.
70. Murray A, Shyy T-K. Integrating attribute and space characteristics in choropleth display and spatial data mining. Int J Geogr Inf Sci. 2000. https://doi.org/10.1080/136588100424954.
71. Boscoe FP, Pickle LW. Choosing geographic units for choropleth rate maps, with an emphasis on public health applications. Cartogr Geogr Inf Sci. 2003;30(3):237–48.
72. Lim LY, Kjellstrom T, Sleigh A, Khamman S, Seubsman SA, Dixon J, et al. Associations between urbanisation and components of the health-risk transition in Thailand: a descriptive study of 87,000 Thai adults. Glob Health Action. 2009;2(1):1914.
73. Kuddus MA, Tynan E, McBryde E. Urbanization: a problem for the rich and the poor? Public Health Rev. 2020;41:1.
74. Morenz AM, Liao JM, Au DH, Hayes SA. Area-level socioeconomic disadvantage and health care spending: a systematic review. JAMA Netw Open. 2024;7(2):e2356121.
75. Trinidad SB, Shalev D, Butcher F, Crawley GK, Ramana S, Clark West R, et al. Use of area-based socioeconomic deprivation indices: a scoping review and qualitative analysis. Health Aff (Millwood). 2022;41(12):1804–11.
76. Rohleder S, Costa DD, Bozorgmehr PK. Area-level socioeconomic deprivation, non-national residency, and COVID-19 incidence: a longitudinal spatiotemporal analysis in Germany. eClin Med. 2022;49:101485.

77. Shenoy DA, Lusk JB, Mahoney H, Blass B, Clark AG, Sloan CE, et al. Associations between neighborhood socioeconomic status, readmission, and mortality for patients with cancer: a nationwide cohort study. Cancer Epidemiol. 2025;98:102907.
78. Singh GK, Siahpush M. Widening socioeconomic inequalities in US life expectancy, 1980–2000. Int J Epidemiol. 2006;35(4):969–79.
79. Robin TA, Khan MA, Kabir N, Rahaman ST, Karim A, Mannan II, et al. Using spatial analysis and GIS to improve planning and resource allocation in a rural district of Bangladesh. BMJ Glob Health. 2019;4:e000832.
80. Qian Y, Qiao G, Li T, Ma J. Measuring the healthcare spatial deprivation in multiple perspectives: a case study of Ningbo city. Front Sustain Cities. 2023;5:1260892.
81. NESDC. Analytical report on poverty and inequality in Thailand 2013. Bangkok: B C Press; 2015.
82. Carstairs V, Morris R. Deprivation: explaining differences in mortality between Scotland and England and Wales. BMJ. 1989;299(6704):886–9.
83. Jarman B. Identification of underprivileged areas. Br Med J (Clin Res Ed). 1983;286(6379):1705–9.
84. Shucksmith M, Roberts D, Scott D. Disadvantage in rural areas: a report to the rural development commission. Aberdeen: University of Aberdeen; 1996.
85. Payne S, Henson B, Gordon D. Poverty and deprivation in West Cornwall in the 1990s. Bristol: Statistical Monitoring Unit, School for Policy Studies, University of Bristol; 1996.
86. Barnett S, Roderick P, Martin D, Diamond I. A multilevel analysis of the effects of rurality and social deprivation on premature limiting long term illness. J Epidemiol Community Health. 2001;55(1):44–51.
87. Adams J, Ryan V, White M. How accurate are Townsend Deprivation Scores as predictors of self-reported health? A comparison with individual level data. J Public Health. 2005;27(1):101–6.
88. Cotgrove AJ, Bell G, Katona CL. Psychiatric admissions and social deprivation: is the Jarman underprivileged area score relevant? J Epidemiol Community Health. 1992;46(3):245–7.
89. Tontisirin N, Anantsuksomsri S. Economic development policies and land use changes in Thailand: from the Eastern Seaboard to the Eastern Economic Corridor. Sustainability. 2021;13:6153.
90. Hutasavi S, Chen D. Exploring the industrial growth and poverty alleviation through space-time data mining from night-time light images: a case study in Eastern Economic Corridor (EEC), Thailand. Int J Remote Sens. 2021;42(14):5271–93.
91. Doungsuwan N, Ratanachai C, Somgpongchaiyakul P, Sangganjanavanich P. Impacts of the national economic and social development plan on Songkhla Lake basin development Thailand. Internat & Econ Res J (IBER). 2013;12(8):895.

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